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Supplementary information

Cortical-like dynamics in recurrent circuits optimized for sampling-based probabilistic inference

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Cortical-like dynamics in recurrent circuits optimized for sampling-based probabilistic inference

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Supplementary Math Note

S1 Mapping from GSM variables to neural responses

Our choice to have membrane potential values rather than firing rates in the network represent the latent variables of the GSM might seem unusual at first, as most computational theories assign meaningful representations to firing rates, not membrane potentials[1] (but see e.g. Ref. 2). However, note that in our case, non-negative firing rates were inappropriate for representing the original GSM variables which could be negative as well as positive. This mismatch could be addressed in either of two mathematically equivalent ways. Membrane potential variables (which can be positive or negative, i.e. above or below the resting value) could represent the GSM variables and firing rates could be computed "phenomenologically" as some transformation of membrane potentials (Eq. 9). Alternatively, we could have defined a transformed generative model in terms of variables that are directly represented by firing rates, with these variables then being appropriately (inversely) transformed into the original GSM variables. Importantly, as we have shown in Ref. 3, these two solutions are mathematically equivalent in that they imply the same predictive distribution over stimuli $\mathcal{P}(\mathbf{x})$ (which determines the model's statistical validity as a model of natural image patches) and ultimately the same target posteriors $\mathcal{P}(\mathbf{r}|\mathbf{x})$ over the rates $\mathbf{r}$ for any stimulus $\mathbf{x}$ (which determine the model's predictions for firing rates). We chose the first solution for two reasons. First, following Ref. 4, the form of network dynamics we used was primarily defined in terms of membrane potentials (Eq. 8), with rates derived "phenomenologically" (Eq. 9). Second, the GSM posteriors were approximately Gaussian, and as such well characterised by their first two moments. This made them suitable targets for the second-order moment-matching objective we used for training our network (Eqs. 25-28). In contrast, the second solution would have required matching highly non-Gaussian target distributions[5].

We also introduced a further transformation, such that even membrane potential values, $\mathbf{u}$, represented a (mildly) transformed version of the GSM latent variables. This was because under the GSM model we used (Eqs. 1-6, Extended Data Fig. 1), the posterior variances over latent variables $\mathbf{y}$ only depended on contrast, but not on anything else in the stimulus. In particular, they did not depend on the orientation content of the stimulus. In contrast, empirical observations from V1 recordings indicate that Fano factors, related to $\mathbf{r}$ and thus ultimately $\mathbf{u}$ in our neural network model (Eqs. 8 and 9), do depend on stimulus orientation[4]. Just as above, this mismatch could have also been addressed in either of two mathematically equivalent ways: via a non-linear transformation of the latent variables $\mathbf{y}$ once they have been inferred, or by modifying the generative model such that it was already defined in terms of these transformed variables $\mathbf{u}$. Once again, these two models are mathematically equivalent in terms of their predictive distribution over image patches and target posterior distributions over membrane potentials, and thus our choice was dictated by practical considerations. For mathematical convenience, we chose the former formalization as there is no *a priori* reason to believe that the brain should employ a linear encoding of the GSM variables. Thus, as described in the Methods, the original GSM latent variables y_i were transformed into membrane potentials according to Eq. 7, with parameters α_{nl}, β_{nl}, and γ_{nl} respectively controlling the scaling, baseline, and power of the transformation. To obtain the kind of variance modulation observed in experiments (variance smallest for neurons driven by their preferred inputs), we set $\gamma_{\mathrm{nl}} < 1$ (Table S1). This also yielded physiologically realistic ranges for the corresponding rates (Fig. 5a). We also note that with the chosen parameters, all the mean values of the original $\mathbf{y}$ distributions lay well above threshold (Fig. S1a), implying only a mild transformation of variables (compare Fig. S1b and c).

S2 Mathematical analyses of oscillations and transients

In order to gain an understanding of the computational benefits of oscillations and transient overshoots in the network, we constructed a series of simplified, analytically tractable systems in which these features could be selectively controlled. In line with our training objective (Eq. 25) and all other analyses presented here, which were based on first- and second-order statistics, we modelled the statistics of neural responses as Gaussian processes (GPs[6]), $\mathbf{u}(t) \sim \mathrm{GP}\left(\boldsymbol{\mu}_{\mathrm{GP}}(\cdot), \boldsymbol{\Sigma}_{\mathrm{GP}}(\cdot,\cdot)\right)$, with mean, $\boldsymbol{\mu}_{\mathrm{GP}}(t) = \mathbb{E}\left[\mathbf{u}(t)\right]$, and covariance functions, $\boldsymbol{\Sigma}_{\mathrm{GP}}\left(t, t'\right) = \mathbb{C}\left[\mathbf{u}(t), \mathbf{u}\left(t'\right)\right]$, chosen to produce overshoots and/or oscillations in a controlled way.

S2.1 Sampling accuracy of statistically stationary dynamics

We want to know how well a finite number of samples (i.e. responses from the network) – or more precisely: a continuum of temporally correlated samples collected over a finite time window of length T – can represent the target distribution $\mathcal{P}$. For this, we quantify the discrepancy between the relevant target distribution, $\mathcal{P}$, and

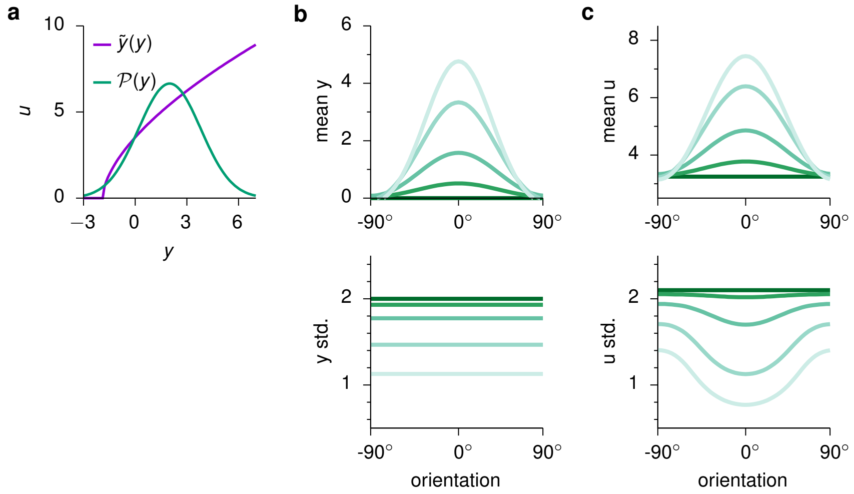

Fig. S1 | Nonlinear transformation of the GSM variables. a, Transformed variable u as a function of the latent variable y of the GSM (purple; Eq. 7), and a representative posterior distribution over y (green). **b-c,** First two moments (top: mean; bottom: standard deviation) of the distributions over the original GSM variables $\mathbf{u}$ (**b**) and the transformed variables $\mathbf{y}$ (**c**). Colors indicate contrast, as in Fig. 2c.

the (time-marginalized) distribution of responses produced by the network over a finite time window T, $\mathcal{Q}_T$. Specifically, we use the symmetrized Kullback-Leibler divergence ($\mathrm{D_{SKL}}$) defined for two probability distributions $\mathcal{P}$ and $\mathcal{Q}_T$ as:

$$\mathrm{D_{SKL}}\left[\mathcal{P}\|\mathcal{Q}_T\right] = \frac{1}{2}\left[\int \mathcal{P}(\mathbf{u}) \log\left(\frac{\mathcal{P}(\mathbf{u})}{\mathcal{Q}_T(\mathbf{u})}\right) \mathrm{d}\mathbf{u} + \int \mathcal{Q}_T(\mathbf{u}) \log\left(\frac{\mathcal{Q}_T(\mathbf{u})}{\mathcal{P}(\mathbf{u})}\right) \mathrm{d}\mathbf{u}\right] \tag{S1}$$

where $\mathrm{D_{SKL}}\left[\mathcal{P}\|\mathcal{Q}_T\right] \geq 0$, with equality achieved if and only if $\mathcal{P}(\mathbf{u}) = \mathcal{Q}_T(\mathbf{u})\ \forall \mathbf{u}$. Note that all pairs of distributions studied here are approximately Gaussian: $\mathcal{P}$ is (approximately) Gaussian given the construction of our generative model (Eqs. 3, 6 and 7), and $\mathcal{Q}_T$ is also Gaussian under our GP approximation (see above). Therefore, both $\mathcal{P}$ and $\mathcal{Q}_T$ can be considered to be fully characterized by their means and covariances: $\boldsymbol{\mu}_\mathcal{P}$ and $\boldsymbol{\Sigma}_\mathcal{P}$ for the target distribution, and $\boldsymbol{\mu}_\mathcal{Q}(T) = \frac{1}{T}\int_0^T \mathbf{u}(t)\,\mathrm{d}t$ and $\boldsymbol{\Sigma}_\mathcal{Q}(T) = \frac{1}{T}\int_0^T \mathbf{u}(t)\ \mathbf{u}^\mathsf{T}(t)\,\mathrm{d}t - \boldsymbol{\mu}_\mathcal{Q}\,\boldsymbol{\mu}_\mathcal{Q}^\mathsf{T}$ for the response distribution, respectively. With these definitions, Eq. S1 takes the following approximate form:

$$\mathrm{D_{SKL}}\left[\mathcal{P}\|\mathcal{Q}_T\right] \approx \frac{1}{4}\left[\boldsymbol{\mathcal{E}}^\mathsf{T}(T)\ \left(\boldsymbol{\Sigma}_\mathcal{P}^{-1} + \boldsymbol{\Sigma}_\mathcal{Q}^{-1}(T)\right)\ \boldsymbol{\mathcal{E}}(T) + \mathrm{Tr}\left(\boldsymbol{\Sigma}_\mathcal{P}^{-1}\,\boldsymbol{\Sigma}_\mathcal{Q}(T)\right) + \mathrm{Tr}\left(\boldsymbol{\Sigma}_\mathcal{Q}^{-1}(T)\ \boldsymbol{\Sigma}_\mathcal{P}\right) - 2\dim(\mathbf{u})\right] \tag{S2}$$

where $\boldsymbol{\mathcal{E}}(T) = \boldsymbol{\mu}_\mathcal{Q}(T) - \boldsymbol{\mu}_\mathcal{P}$ is the error in the sample mean of responses.

We now consider the special case when $\dim(\mathbf{u}) = 1$, i.e. $\mathbf{u}$ is a scalar u (either a single cell, or an arbitrary 1-dimensional projection of the network's activity – see Footnote 2 at the end of this section as well as Sections S2.2.2–S2.2.4 for results pertaining to multivariate $\mathbf{u}$), and thus so are the corresponding means and variances in Eq. S2. Note that $\mu_\mathcal{Q}(T)$ and $\sigma^2_\mathcal{Q}(T)$ themselves are random variables (as they are based on the stochastic responses, $u(t)$, of the network). Thus, we compute the *expected* divergence between $\mathcal{P}$ and $\mathcal{Q}_T$, $\overline{\mathrm{D}}_\mathrm{SKL}\left[\mathcal{P}\|\mathcal{Q}_T\right] = \mathbb{E}\left[\mathrm{D_{SKL}}\left[\mathcal{P}\|\mathcal{Q}_T\right]\right]$, over the distribution of $\mu_\mathcal{Q}(T)$ and $\sigma^2_\mathcal{Q}(T)$, given the "true" mean and covariance function of the GP, $\mu_\mathrm{GP}(t)$ and $\sigma^2_\mathrm{GP}\left(t, t'\right)$, producing the samples comprising $\mathcal{Q}_T$:

$$\overline{\mathrm{D}}_\mathrm{SKL}\left[\mathcal{P}\|\mathcal{Q}_T\right] \approx \frac{1}{4}\left[\mathbb{E}\left[\mathcal{E}^2(T)\ \left(\frac{1}{\sigma^2_\mathcal{P}} + \frac{1}{\sigma^2_\mathcal{Q}(T)}\right)\right] + \mathbb{E}\left[\sigma^2_\mathcal{Q}(T)\right]\frac{1}{\sigma^2_\mathcal{P}} + \mathbb{E}\left[\frac{1}{\sigma^2_\mathcal{Q}(T)}\right]\ \sigma^2_\mathcal{P} - 2\right] \tag{S3}$$

Here we study the dynamics of our network in the statistically stationary limit, i.e. past burn-in (see Section S2.3 for an analysis of transients). Formally, this means that $\mathbb{E}\left[u(t)\right] = \mu_\mathrm{GP}(t) = \mu_\mathrm{GP}$, $\mathbb{V}\left[u(t)\right] = \sigma^2_\mathrm{GP}(t,t) = \sigma^2_\mathrm{GP}$, and $\mathbb{C}\left[u(t), u\left(t'\right)\right] = \sigma^2_\mathrm{GP}\ C_\mathrm{GP}\left(t - t'\right)\ \ \forall t, t'$, where $C_\mathrm{GP}(\Delta t) = C_\mathrm{GP}(-\Delta t)$ is the autocorrelation function of the GP characterizing network responses. As we are primarily interested in how sampling accuracy specifically depends on the *dynamics* of neural responses, we also assume that the (time-marginalized) stationary distribution of responses has already been matched to the target distribution. This implies that $\mu_{\mathcal{P}} = \mathbb{E}\left[\mu_{\mathcal{Q}}(T)\right] = \mu_{\mathrm{GP}}$ and $\sigma^2_{\mathcal{P}} = \mathbb{E}\left[\sigma^2_{\mathcal{Q}}(T)\right] = \sigma^2_{\mathrm{GP}}\left(1 - \overline{C}_{\mathrm{GP}}(T)\right)$, where

$$\overline{C}_{\mathrm{GP}}(T) = \frac{1}{T^2}\int_0^T\int_0^T C_{\mathrm{GP}}\left(t - t'\right)\mathrm{d}t\,\mathrm{d}t' \tag{S4}$$

is the average autocorrelation of the network, i.e. the normalized area under the autocorrelation function. Therefore, note that moment matching does not constrain C_{GP} in any way, as any value of $\overline{C}_{\mathrm{GP}}(T)$ (smaller than 1) can be compensated for by the appropriate choice of σ^2_{GP} to match $\mathbb{E}\left[\sigma^2_{\mathcal{Q}}(T)\right]$ to $\sigma^2_{\mathcal{P}}$. For simplicity, we further assume that the sample mean, $\mu_{\mathcal{Q}}(T)$, and variance of network responses, $\sigma^2_{\mathcal{Q}}(T)$, are statistically independent given μ_{GP}, σ^2_{GP} and C_{GP}. (Formally, this only holds for i.i.d. normal random variables, while for correlated variables produced by a stationary process, such as our sequence of network responses, only zero covariance can be proven.[1]) In addition, we ignore the effects of variability in the sample variance, $\sigma^2_{\mathcal{Q}}(T)$. Under these assumptions Eq. S3 (or at least a lower bound thereof) can be written as

$$\overline{\mathrm{D}}_{\mathrm{SKL}}\left[\mathcal{P}\|\mathcal{Q}_T\right] \approx \frac{\overline{\mathcal{E}}^2(T)}{2\,\sigma^2_{\mathcal{P}}} \tag{S5}$$

where $\overline{\mathcal{E}}^2(T) = \mathbb{E}\left[\mathcal{E}^2(T)\right] = \mathbb{V}\left[\mu_{\mathcal{Q}}(T)\right]$ is the mean squared error of the sample mean of responses, which is equal to the variance of the sample mean under our moment matching assumption (see above). Importantly, in this case, $\overline{\mathcal{E}}^2(T)$ can be shown to depend directly on the average autocorrelation of network responses:

$$\overline{\mathcal{E}}^2(T) = \sigma^2_{\mathcal{P}}\,\frac{\overline{C}_{\mathrm{GP}}(T)}{1 - \overline{C}_{\mathrm{GP}}(T)} \tag{S6}$$

Substituting Eq. S6 to Eq. S5 yields[2]

$$\overline{\mathrm{D}}_{\mathrm{SKL}}\left[\mathcal{P}\|\mathcal{Q}_T\right] \approx \frac{1}{2}\,\frac{\overline{C}_{\mathrm{GP}}(T)}{1 - \overline{C}_{\mathrm{GP}}(T)} \tag{S7}$$

As the area under the autocorrelogram must be finite, Eqs. S4 and S7 imply that $\overline{C}_{\mathrm{GP}}(T)$ and thus $\overline{\mathrm{D}}_{\mathrm{SKL}}$ tends to 0 as $T \to \infty$. In other words, in line with intuition, sufficiently long time windows allow the target distribution to be represented with arbitrary precision (once the stationary distribution is matched).

S2.2 Oscillations

In this section we first analyze how oscillations in neural responses affect sampling accuracy in general, and then study how the dynamics of a network lead to such oscillations, and in particular to oscillations that are beneficial for sampling.

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[1] Alternatively, the results below can also be obtained without assuming the independence of the sample mean and variance, by using the expected (non-symmetrized) Kullback-Leibler divergence, $\overline{\mathrm{D}}_{\mathrm{KL}}\left(\mathcal{Q}\|\mathcal{P}\right)$, to measure the sampling accuracy of the network.

[2] Eq. S7 can be straightforwardly generalized to the original case of multivariate $\mathbf{u}$:

$$\overline{\mathrm{D}}_{\mathrm{SKL}}\left[\mathcal{P}\|\mathcal{Q}_T\right] \approx \frac{1}{2}\mathrm{Tr}\left[\left(\mathbf{I} - \overline{\mathbf{C}}_{\mathrm{GP}}(T)\right)^{-1}\overline{\mathbf{C}}_{\mathrm{GP}}(T)\right]$$

where, in analogy with Eq. S4,

$$\overline{\mathbf{C}}_{\mathrm{GP}}(T) = \frac{1}{T^2}\int_0^T\int_0^T \mathbf{C}_{\mathrm{GP}}\left(t - t'\right)\mathrm{d}t\,\mathrm{d}t'$$

such that

$$\mathbb{C}\left[\mathbf{u}(t), \mathbf{u}\left(t'\right)\right] = \mathbf{\Sigma}^{1/2}_{\mathrm{GP}}\,\mathbf{C}_{\mathrm{GP}}\left(t - t'\right)\,\mathbf{\Sigma}^{1/2}_{\mathrm{GP}} \quad \text{with} \quad \mathbf{C}_{\mathrm{GP}}(-\Delta t) = \mathbf{C}^{\mathrm{T}}_{\mathrm{GP}}(\Delta t) \quad \text{and} \quad \mathbf{C}_{\mathrm{GP}}(0) = \mathbf{I}$$

in the stationary case.

S2.2.1 Oscillations and sampling accuracy

The fact that there is a tight relationship between the average autocorrelation and sampling accuracy (Eq. S7) reveals a fundamental benefit of oscillations for sampling. Oscillations make the total (signed) area under the autocorrelogram smaller because an oscillatory autocorrelogram has as its envelope (and thus never exceeds) its non-oscillatory counterpart (Eq. 35). In particular, with the parametrization of the autocorrelogram used in the main text (Eq. 36), it is easy to show that $\overline{\mathrm{D}}_{\mathrm{SKL}}\left[\mathcal{P}\|\mathcal{Q}\right] \to 0$ in Eq. S7 (because $\overline{C}_{\mathrm{GP}}(T) \to 0$) as $f \to \infty$, provided that $\chi > 0$. Conversely, the more oscillatory the system is (i.e. the larger χ is), the better its accuracy (the smaller its $\overline{\mathrm{D}}_{\mathrm{SKL}}\left[\mathcal{P}\|\mathcal{Q}\right]$) becomes for fixed, sufficiently high f.

S2.2.2 Oscillatoriness of high-dimensional network dynamics

In the preceding sections, we have used GPs with oscillatory autocorrelations as phenomenological models in order to understand how oscillatory components in network responses confer benefits for sampling. In this section, we build analytical insight into how the dynamics of the actual network gives rise to its spatio-temporal covariance, and its oscillatoriness, in the first place.

The first two moments of the stationary distribution of network activity are given by the steady-state solutions of the differential equations in Eqs. 18-20:

$$\mathbf{0} = \left[\mathbf{T}^{-1}\,\mathbf{\Sigma}^*\right] + \left[\mathbf{T}^{-1}\,\mathbf{\Sigma}^*\right]^{\mathsf{T}} + \mathcal{J}\,\mathbf{\Sigma} + \mathbf{\Sigma}\,\mathcal{J}^{\mathsf{T}} \tag{S8}$$

$$\mathbf{0} = -\frac{1}{\tau_\eta}\,\mathbf{\Sigma}^* + \mathbf{\Sigma}^\eta\,\mathbf{T}^{-1} + \mathbf{\Sigma}^*\,\mathcal{J}^{\mathsf{T}} \tag{S9}$$

$$\frac{\mathrm{d}\mathbf{\Sigma}(\Delta t)}{\mathrm{d}\Delta t} = e^{-\Delta t/\tau_\eta}\,\left[\mathbf{T}^{-1}\mathbf{\Sigma}^*\right]^{\mathsf{T}} + \mathbf{\Sigma}(\Delta t)\,\mathcal{J}^{\mathsf{T}}, \qquad \forall \Delta t > 0 \tag{S10}$$

where we have dropped the primary time index t everywhere as we are studying the stationary distribution here. The case $\Delta t < 0$ in Eq. S10 is resolved using $\mathbf{\Sigma}(-\Delta t) = \mathbf{\Sigma}^{\mathsf{T}}(\Delta t)$. Eq. S9 can be solved directly:

$$\mathbf{\Sigma}^* = \mathbf{\Sigma}^\eta\,\mathbf{T}^{-1}\,\left(\mathbf{I}/\tau_\eta - \mathcal{J}^{\mathsf{T}}\right)^{-1} \tag{S11}$$

and so Eq. S10 can be rewritten for convenience as:

$$\frac{\mathrm{d}\mathbf{\Sigma}^{\mathsf{T}}(\Delta t)}{\mathrm{d}\Delta t} = e^{-\Delta t/\tau_\eta}\,\mathbf{M} + \mathcal{J}\,\mathbf{\Sigma}^{\mathsf{T}}(\Delta t)\,, \quad \text{where} \quad \mathbf{M} = \mathbf{T}^{-1}\,\mathbf{\Sigma}^* = \mathbf{T}^{-1}\,\mathbf{\Sigma}^\eta\,\mathbf{T}^{-1}\,\left(\mathbf{I}/\tau_\eta - \mathcal{J}^{\mathsf{T}}\right)^{-1} \tag{S12}$$

Now, we use the Laplace transform to obtain:

$$\mathcal{L}\left\{\mathbf{\Sigma}^{\mathsf{T}}\right\}(s) = (s\,\mathbf{I} - \mathcal{J})^{-1}\,\left[\mathbf{\Sigma}^{\mathsf{T}}(0) + \frac{1}{s + 1/\tau_\eta}\mathbf{M}\right] \tag{S13}$$

which is then solved by using the inverse Laplace transform:

$$\mathbf{\Sigma}^{\mathsf{T}}(\Delta t) = e^{\Delta t\,\mathcal{J}} * \left[\delta(\Delta t)\,\mathbf{\Sigma}^{\mathsf{T}}(0) + e^{-\Delta t/\tau_\eta}\,\mathrm{H}(\Delta t)\,\mathbf{M}\right] \tag{S14}$$

where $\mathrm{H}(\cdot)$ is the Heaviside function. This yields:

$$\mathbf{\Sigma}^{\mathsf{T}}(\Delta t) = e^{\Delta t\,\mathcal{J}}\,\mathbf{\Sigma}^{\mathsf{T}}(0) + \left(\mathbf{I}/\tau_\eta + \mathcal{J}\right)^{-1}\,\left[e^{\mathcal{J}\,\Delta t} - e^{-\Delta t\,\mathbf{I}/\tau_\eta}\right]\,\mathbf{M} \tag{S15}$$

and therefore

$$\mathbf{\Sigma}(\Delta t) = \mathbf{\Sigma}\,e^{\Delta t\,\mathcal{J}^{\mathsf{T}}} + \mathbf{\Sigma}^{*\mathsf{T}}\,\mathbf{T}^{-1}\,\left[e^{\Delta t\,\mathcal{J}^{\mathsf{T}}} - e^{-\Delta t\,\mathbf{I}/\tau_\eta}\right]\,\left(\mathbf{I}/\tau_\eta + \mathcal{J}^{\mathsf{T}}\right)^{-1} \tag{S16}$$

Finally, by diagonalizing the Jacobian $\mathcal{J} = \mathbf{V}\,\Lambda\,\mathbf{V}^{-1}$, where Λ is a diagonal matrix of eigenvalues and $\mathbf{V}$ is the matrix whose columns contain the eigenvectors of $\mathcal{J}$, one obtains:

$$\mathbf{\Sigma}(\Delta t) = \mathbf{\Sigma}\,\mathbf{V}^{-\mathsf{T}}\,e^{\Delta t\,\Lambda}\,\mathbf{V}^{\mathsf{T}} + \mathbf{\Sigma}^{*\mathsf{T}}\,\mathbf{T}^{-1}\,\mathbf{V}^{-\mathsf{T}}\,\left[e^{\Delta t\,\Lambda} - e^{-\Delta t\,\mathbf{I}/\tau_\eta}\right]\,\left(\mathbf{I}/\tau_\eta + \Lambda\right)^{-1}\,\mathbf{V}^{\mathsf{T}} \tag{S17}$$

Eq. S17 exposes how the eigenvalues of the Jacobian (diagonal of Λ) with non-zero imaginary parts (first term of the equation and first term in square brackets) contribute to oscillatoriness in the sense of Eqs. 33 and 36. (Oscillations in the correlogram may be damped if the corresponding eigenvalues have comparatively large negative real parts.) This complex component competes with a pure (real) exponential decay (second term in square brackets), reflecting the autocorrelation of process noise. Thus, overall, Eq. S17 results in a partial degree of oscillatoriness (modulated by properties of the Jacobian), i.e. $\chi < 1$, as seen in the real network.

S2.2.3 Optimal oscillatory high-dimensional dynamics

To understand how oscillatory activity in high-dimensional dynamics can be optimised for sampling, we use (an approximation of) our original measure of accuracy for multivariate distributions, the multivariate equivalent of Eq. S5:

$$\overline{\mathrm{D}}_{\mathrm{SKL}}\left[\mathcal{P}\|\mathcal{Q}_T\right] \approx \frac{1}{2}\mathbb{E}\left[\boldsymbol{\mathcal{E}}^{\mathsf{T}}(T)\,\boldsymbol{\Sigma}_{\mathcal{P}}^{-1}\,\boldsymbol{\mathcal{E}}(T)\right] \tag{S18}$$

Using the equivalence of a scalar expression and its own trace, and the cyclic property of the trace, this can be rewritten as

$$\overline{\mathrm{D}}_{\mathrm{SKL}}\left[\mathcal{P}\|\mathcal{Q}_T\right] \approx \frac{1}{2}\mathrm{Tr}\left[\mathbb{C}\left[\boldsymbol{\mu}_{\mathcal{Q}}(T)\right]\,\boldsymbol{\Sigma}_{\mathcal{P}}^{-1}\right] \tag{S19}$$

Here we extend the results of Ref. 7 on a network receiving fast (temporally white) process noise, to the case of temporally correlated noise, which is more relevant to our optimized network. For analytical tractability, we have made two simplifying assumptions regarding process noise: that its temporal correlations are short (though non-negligible) compared to the intrinsic time constants of the network, and that it is spatially white (see below for mathematical definitions). Moreover, we assume that the dynamics of the inhibitory neurons are fast enough that their responses become slaved to those of the E neurons. This implies that we can describe our network with effective dynamics in the E population alone, i.e. the same space over which the target distributions are defined. In turn, this will help establish an analytical link between network dynamics and sampling performance.

In the following, we assume stationarity and fast process noise, $\tau_\eta\,\mathcal{J} \ll \mathbf{I}$, expanding expressions to first order in τ_η, and define $\mathbf{Q}_\circ = 2\,\tau_\eta\,\mathbf{T}^{-1}\,\boldsymbol{\Sigma}^{\eta}\,\mathbf{T}^{-1}$. With these assumptions and definition, after substituting Eq. S11 into Eq. S16, we can rewrite the lagged-covariance of the network as:

$$\boldsymbol{\Sigma}(\Delta t) \approx \left(\boldsymbol{\Sigma} + \frac{\tau_\eta}{2}\,\mathbf{Q}_\circ\right)\,e^{\Delta t \mathcal{J}^{\mathsf{T}}} - \frac{\tau_\eta}{2}\,\mathbf{Q}_\circ\,e^{-\Delta t/\tau_\eta} \tag{S20}$$

Under our assumptions, we can also establish a relationship between $\mathcal{J}$ and $\boldsymbol{\Sigma}$ based on Eq. S8, via the following Lyapunov equation:

$$0 \approx \mathbf{Q}_\circ + \mathcal{J}\left(\boldsymbol{\Sigma} + \frac{\tau_\eta}{2}\,\mathbf{Q}_\circ\right) + \left(\boldsymbol{\Sigma} + \frac{\tau_\eta}{2}\,\mathbf{Q}_\circ\right)\mathcal{J}^{\mathsf{T}} \tag{S21}$$

Analogously to the case of temporally white process noise[7], we can parameterize any solution to the Lyapunov equation Eq. S21 as:

$$\mathcal{J} = -\left(\frac{1}{2}\,\mathbf{Q}_\circ + \mathbf{S}\right)\left(\boldsymbol{\Sigma} + \frac{\tau_\eta}{2}\,\mathbf{Q}_\circ\right)^{-1} \tag{S22}$$

where $\mathbf{S}$ is an arbitrary skew-symmetric matrix, which means that it can be diagonalized as

$$\mathbf{S} = \mathbf{U}\,\boldsymbol{\Omega}^{-1}\,\mathbf{U}^{\star} \tag{S23}$$

where $\mathbf{U}$ is a (complex) unitary matrix of eigenvectors, $\mathbf{U}^{\star}$ denotes its conjugate transpose, and $\boldsymbol{\Omega}$ is a diagonal matrix containing the inverses of the purely imaginary (or zero) eigenvalues of $\mathbf{S}$.

Eqs. S20 and S22 allow us to rewrite the covariance of $\boldsymbol{\mu}_{\mathcal{Q}}(T)$, the quantity relevant for $\overline{\mathrm{D}}_{\mathrm{SKL}}\left[\mathcal{P}\|\mathcal{Q}_T\right]$ (Eq. S19), as:

$$\mathbb{C}\left[\boldsymbol{\mu}_{\mathcal{Q}}(T)\right] = \frac{1}{T^2}\int_0^T\!\!\int_0^T \boldsymbol{\Sigma}(t-t')\,\mathrm{d}t\,\mathrm{d}t' \tag{S24}$$

$$\approx \frac{1}{T}\left[\left(\boldsymbol{\Sigma} + \frac{\tau_\eta}{2}\,\mathbf{Q}_\circ\right)\left(\frac{1}{2}\,\mathbf{Q}_\circ - \mathbf{S}\right)^{-1}\mathbf{Q}_\circ\left(\frac{1}{2}\,\mathbf{Q}_\circ + \mathbf{S}\right)^{-1}\left(\boldsymbol{\Sigma} + \frac{\tau_\eta}{2}\,\mathbf{Q}_\circ\right)\right] \tag{S25}$$

where we have assumed large T. To simplify the analysis, we follow Ref. 7 and set $\boldsymbol{\Sigma}_\eta = \sigma^2\,\mathbf{I}$ and $\mathbf{Q}_\circ = 2\,\frac{\tau_\eta\,\sigma^2}{\tau^2}\,\mathbf{I}$. We thus have:

$$\mathbb{C}\left[\boldsymbol{\mu}_{\mathcal{Q}}(T)\right] \approx \frac{2\,\tau_\eta\,\sigma^2}{\tau^2\,T}\left[\left(\boldsymbol{\Sigma} + \frac{\tau_\eta^2\,\sigma^2}{\tau^2}\,\mathbf{I}\right)\left(\frac{\tau_\eta\,\sigma^2}{\tau^2}\,\mathbf{I} - \mathbf{S}\right)^{-1}\left(\frac{\tau_\eta\,\sigma^2}{\tau^2}\,\mathbf{I} + \mathbf{S}\right)^{-1}\left(\boldsymbol{\Sigma} + \frac{\tau_\eta^2\,\sigma^2}{\tau^2}\,\mathbf{I}\right)\right] \tag{S26}$$

which, to first order in τ_η, reduces to

$$\mathbb{C}\left[\boldsymbol{\mu}_{\mathcal{Q}}(T)\right] \approx -\frac{2\,\tau_\eta\,\sigma^2}{\tau^2\,T}\,\boldsymbol{\Sigma}\,\mathbf{S}^{-2}\,\boldsymbol{\Sigma} \tag{S27}$$

For the moment matched case, $\boldsymbol{\Sigma}_{\mathcal{P}} = \boldsymbol{\Sigma}_{\mathcal{Q}} = \boldsymbol{\Sigma}^{1/2}\left(\mathbf{I} - \overline{\mathbf{C}}(T)\right)\boldsymbol{\Sigma}^{1/2} \approx \boldsymbol{\Sigma}$ (for sufficiently large T), which means

$$\mathbb{C}\left[\boldsymbol{\mu}_{\mathcal{Q}}(T)\right] \approx -\frac{2\,\tau_\eta\,\sigma^2}{\tau^2\,T}\,\boldsymbol{\Sigma}_{\mathcal{P}}\,\mathbf{S}^{-2}\,\boldsymbol{\Sigma}_{\mathcal{P}} \tag{S28}$$

We then substitute Eq. S28 into Eq. S19 to obtain

$$\overline{\mathrm{D}}_{\mathsf{SKL}}\left[\mathcal{P}\|\mathcal{Q}_T\right] \approx -\frac{\tau_\eta\,\sigma^2}{\tau^2\,T}\mathrm{Tr}\left[\boldsymbol{\Sigma}_{\mathcal{P}}\,\mathbf{S}^{-2}\right] \tag{S29}$$

Therefore, optimizing the network amounts to finding the $\mathbf{S}$ that minimizes the sampling error, $\overline{\mathrm{D}}_{\mathsf{SKL}}\left[\mathcal{P}\|\mathcal{Q}_T\right]$, given by Eq. S29. It is immediately clear that increasing the scale of $\mathbf{S}$ trivially decreases $\overline{\mathrm{D}}_{\mathsf{SKL}}\left[\mathcal{P}\|\mathcal{Q}_T\right]$. We thus ask what $\mathbf{S}$ achieves a set level of the sampling error, $\overline{\mathrm{D}}^*_{\mathsf{SKL}}$, with the minimal scale (determinant)? The optimal $\mathbf{S}$ can be obtained by solving the following optimization problem:

$$\underset{\mathbf{S}}{\operatorname{argmin}}\,\det(\mathbf{S}) \text{ s.t. } \overline{\mathrm{D}}_{\mathsf{SKL}}\left[\mathcal{P}\|\mathcal{Q}_T\right] = \overline{\mathrm{D}}^*_{\mathsf{SKL}} \tag{S30}$$

The solution to this optimization problem is:

$$\mathbf{S}^2 = -\frac{\tau_\eta\,\sigma^2\,N}{\tau^2\,T\,\overline{\mathrm{D}}^*_{\mathsf{SKL}}}\,\boldsymbol{\Sigma}_{\mathcal{P}} \tag{S31}$$

Finally, for the moment matched case we are studying here and assuming fast process noise[3], Eq. S22 can be rewritten as

$$\mathcal{J} \approx -\mathbf{S}\,\boldsymbol{\Sigma}_{\mathcal{P}}^{-1} \tag{S32}$$

Combining Eq. S32 with Eq. S31 and Eq. S23 allows us to express the Jacobian of the network as

$$\mathcal{J} \propto \mathbf{S}^{-1} = \mathbf{U}\,\boldsymbol{\Omega}\,\mathbf{U}^\star \tag{S33}$$

which is also a skew-symmetric matrix with purely imaginary (or zero) eigenvalues given by $\boldsymbol{\Omega}$ (though see Footnote 3). Thus, the optimal dynamics are rotational, resulting in oscillatory responses, with frequencies determined by the magnitude of the elements of $\boldsymbol{\Omega}$. Moreover, from Eqs. S23 and S31, we can derive a relationship between $\boldsymbol{\Omega}$ and $\boldsymbol{\Sigma}_{\mathcal{P}}$:

$$\det(\boldsymbol{\Sigma}_{\mathcal{P}}) \propto \det\left(\mathbf{S}^2\right) = \det{}^{-1}\left(\boldsymbol{\Omega}^2\right) \tag{S34}$$

This means that, overall, oscillation frequencies in the network should scale inversely with the target covariance. Specifically, at progressively higher contrast levels, as $\boldsymbol{\Sigma}_{\mathcal{P}}$ becomes smaller, this relationship predicts higher oscillation frequencies. This is exactly what we observe in the network (Fig. 5c).

This analysis also reveals another interesting property of the optimal dynamics. As $\mathcal{J}$ is skew-symmetric (Eq. S33 and footnote 3), not only does it have purely imaginary (or zero) eigenvalues, as we saw above, but its non-zero eigenvalues also come in positive /negative pairs of equal magnitude, with each pair defining a two-dimensional plane in which the dynamics are oscillatory. Thus, without loss of generality, in the following, we analyze the dynamics by focusing on one such plane at a time.

S2.2.4 Two-dimensional oscillatory dynamics

As seen in the previous section, optimal sampling encourages the network to operate as a collection of two-dimensional oscillators. To understand how the degree of oscillatoriness and the output variance depend on

[3] Note that our approximation to $\mathcal{J}$ is only expanding to 0th order in τ_η, rather than 1st order as until now – otherwise we would have an additional $-\frac{1}{2}\,\mathbf{Q}_\circ\,\boldsymbol{\Sigma}_{\mathcal{P}}^{-1}$ term in Eq. S32. This explains why we get purely oscillatory dynamics rather than partial oscillatoriness as derived in Section S2.2.2 and seen in the real network. Nevertheless, in this section our focus is on understanding specifically the oscillatory components of the dynamics, and so we work with this approximation here.

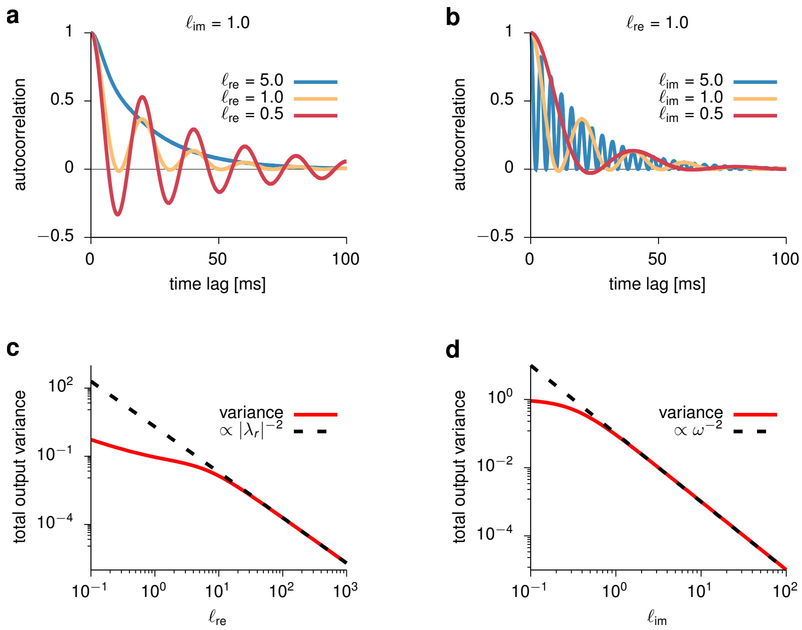

Fig. S2 | Effects of the eigenvalues of the Jacobian in a two-dimensional system on response autocorrelation and variance. a, Autocorrelation as a function of time lag, for different values of ℓ_{re} and constant $\ell_{im} = 1$ (see text following Eq. S35 for definitions). **b,** Same as **a**, but for different values of ℓ_{im} and constant $\ell_{re} = 1$. **c,** Total output variance as a function of ℓ_{re} for constant $\ell_{im} = 1$. **d,** Same as **c**, but as a function of ℓ_{im} for constant $\ell_{re} = 1$.

the eigenvalues of the Jacobian, we analyze here a reduced two-dimensional system, representing one of these planes in the full N-dimensional system. Each plane supports oscillations of a single frequency, given by the imaginary part of the corresponding eigenvalue (λ_{im}). Without loss of generality, we express the Jacobian as

$$\mathcal{J} = \begin{bmatrix} \lambda_{re} & -\lambda_{im} \\ \lambda_{im} & \lambda_{re} \end{bmatrix} \tag{S35}$$

with $\lambda_{re} < 0$ for stability. The eigenvalues of $\mathcal{J}$ are then given by $\lambda = \lambda_{re} \pm i\lambda_{im}$. Let us denote $\ell_{re} = -\lambda_{re}\,\tau_\eta$, and $\ell_{im} = \lambda_{im}\,\tau_\eta/(2\pi)$. To understand the effects of these eigenvalues on network oscillations, we used Eqs. S8, S11 and S17 to compute $\mathbf{\Sigma}^*$ and $\mathbf{\Sigma}$ and the autocorrelation function of the network (Fig. S2). We found that while ℓ_{re} controls the degree of oscillatoriness of the system, ℓ_{im} has little to no effect on it (and instead controls the oscillation frequency as expected; compare Fig. S2a-b). Note, however, that ℓ_{re} does not control the oscillatoriness of the system quite as independently from the envelope of decay in the same way as χ did it in our parametric fit (Eqs. 33 and 36).

It is also instructive to analyze how the total output variance is related to the real and imaginary parts of the Jacobian. In particular, the total variance (given by the trace of $\mathbf{\Sigma}$) decreases with ℓ_{re}, which in turns implies reduced oscillatoriness (Fig. S2c). This predicts a positive correlation between oscillatoriness and variance: larger variance should be associated with small ℓ_{re} and thus with a larger degree of oscillatoriness. This is consistent with what we observed across the different PCs in the full network (Fig. 6c-d). Moreover, the total variance also decays with the square of the oscillation frequency $\omega = \lambda_{im}$ (Fig. S2d), corroborating our earlier explanation of why oscillation frequencies tend to increase with contrast (as output variance decreases).

S2.3 Transients

Any system performing inference via sampling in real time faces the challenge of approximating a time-varying posterior using samples over a finite time window. In the main text, we showed that transient overshoots are useful in this setting. To explain this effect, here we analyzed the related (albeit simpler) task of generating (one dimensional[4]) time-varying neural activity $u(t)$ whose running average, $\mu_{\mathcal{Q}}(t)$[5], most closely tracked a time-varying target signal $\mu_{\mathcal{P}}(t)$ (e.g. the posterior mean in a continual inference scenario). We took the running average to be the convolution of $u(t)$ with a kernel $k(t)$:

$$\mu_{\mathcal{Q}}(t) = \left(u * k\right)(t)\,. \tag{S36}$$

More specifically, our goal was to find the $u(t)$ that minimized:

$$\tilde{\mathrm{D}} = \int \left[\left(\mu_{\mathcal{Q}}(t) - \mu_{\mathcal{P}}(t)\right)^2 + \epsilon_{\text{smooth}} \left|u'(t)\right|^2 \right] \mathrm{d}t \tag{S37}$$

where the first term penalized the average squared distance between $\mu_{\mathcal{Q}}(t)$ and the time-varying target $\mu_{\mathcal{P}}(t)$ (and as such, it was closely related to $\overline{\mathrm{D}}^{\mu}_{\mathrm{SKL}}$, Eq. 38), and the second term encouraged temporal smoothness by penalizing high (squared) temporal derivatives, $u'(t)$, such that the optimal $u(t)$ could have been produced by a continuous dynamical system. By taking the Fourier transform of Eq. S37, and making use of Plancherel's theorem, we obtain:

$$\tilde{\mathrm{D}} = \int \left(\left| \hat{u}(f)\ \hat{k}(f) - \hat{\mu}_{\mathcal{P}}(f) \right|^2 + \epsilon_{\text{smooth}}\ \left(2\pi f\right)^2\ \left|\hat{u}(f)\right|^2 \right) \mathrm{d}f \tag{S38}$$

where $\hat{\cdot}$ denotes the Fourier transform. Setting $\frac{\mathrm{d}\tilde{\mathcal{D}}}{\mathrm{d}u}$ to zero, we obtain the optimal $\hat{u}$:

$$\hat{u}(f) = \frac{\hat{\mu}_{\mathcal{P}}(f)\ \hat{k}^*(f)}{\left|\hat{k}(f)\right|^2 + 4\,\epsilon_{\text{smooth}}\,\pi^2 f^2}, \tag{S39}$$

where $\cdot^*$ denotes the complex conjugate. The optimal $u(t)$ can then be obtained from Eq. S39 via the inverse Fourier transform (Extended Data Fig. 9c).

S3 Online repositories

Code and parameters for the numerical experiments: bitbucket.org/RSE_1987/ssn_inference_numerical_experiments
Code for the optimization procedure: bitbucket.org/RSE_1987/ssn_inference_optimizer

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[4]As before, the results generalize to multidimensional responses.

[5]Note that here we have slightly overloaded our previous notation, as t in $\mu_{\mathcal{Q}}(t)$ refers to the *point in time* up until which we average responses, rather than the *length of time* over which we average, as T did in the similar notation $\mu_{\mathcal{Q}}(T)$ used in Sections S2.1 and S2.2. The duration of the averaging window here is implicit in the definition of $k(t)$.

Parameter	Description	Value
	Generative model	
N_x	Number of observed variables (pixels)	256
N_y	Number of latent variables	50
$\mathbf{A}$	Matrix of Gabor oriented filters	(Extended Data Fig. 1 and repository)
$\mathbf{C}$	Prior covariance matrix	(Extended Data Fig. 1 and repository)
σ_x	Standard deviation of the pixel noise	10.0
K	Shape parameter of the gamma prior on contrast	2.0
ϑ	Scale parameter of the gamma prior on contrast	2.0
α_{nl}	Scaling of the non-linear transformation	2.4
β_{nl}	Baseline of the non-linear transformation	1.9
γ_{nl}	Power of the non-linear transformation	0.6
	Network	
N	Total number of neurons	100
N_E	Number of E cells	50
N_I	Number of I cells	50
τ_E	Membrane time constant of E cells	20ms
τ_I	Membrane time constant of I cells	10ms
τ_η	Process noise time constant	20ms
$\mathbf{W}$	Weight matrix	(optimized, see repository)
n	Exponent in the single cell transfer function	2
k	Scaling factor in the single cell transfer function	0.3
$\mathbf{\Sigma}_\eta$	Process noise covariance	(optimized, see repository)
μ_0	Mean of the initial distribution	$\mathbf{0}$
$\mathbf{\Sigma}_0$	Covariance of the initial distribution	$4\,\mathbf{I}$
dt	simulation time-step	0.2 ms
	Input	
$\mathbf{W}^{ff}$	Feed-forward weights	$[\mathbf{A}\,\mathbf{A}]^T/15.0$
α_h	Input scaling	1.96 (optimized)
β_h	Input baseline	0.10 (optimized)
γ_h	Input power	2.03 (optimized)
	Fano factor calculation	
T	Observation window	100 ms
K_{ISI}	Shape parameter of interspike interval gamma distribution	1.15
	Input conductance calculation	
C_m	Membrane capacitance	20.0 pF
V_{rest}	Resting membrane potential (reference level for $\mathbf{u}$)	-65 mV
V^E	E reversal potential	0 mV
V^I	I reversal potential	-80 mV
	Optimization	
ϵ_{mean}	Cost weight for the error of the mean	$4.0 \cdot 10^{-5}$
ϵ_{var}	Cost weight for the error of the variance	$8.0 \cdot 10^{-5}$
ϵ_{cov}	Cost weight for the error of the covariance	$8.0 \cdot 10^{-7}$
ϵ_{slow}	Cost weight for the slowness penalty	$4.0 \cdot 10^{-7}$ (ADF only)
T_{max}	Stimulus presentation time/ integration end time	500 ms
T_{min}	Integration start time	0 to 450 ms
τ_{max}	Autocorrelation integration time	100 ms
N_{trials}	Number of trials for stochastic optimization	50
dt'	simulation time-step during optimization	0.2 ms

Table S1 | List of parameter values.